

GUESS Framework Practice — Mechanics, Waves, Gases, Radiation & Circuits

The GUESS Framework

For every question, show your working in this order:

G — Given: list the quantities the question gives you, with units.

U — Unknown: state clearly what you are solving for.

E — Equation: write relevant equation from the bank below, in symbols, before touching numbers.

S — Substitute: put the numbers in, keeping units.

S — Solve: rearrange and calculate, finishing with a boxed answer with correct units and sig figs.

Equation Bank

Mechanics

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$F = ma$$

$$p = mv$$

$$F = \frac{\Delta p}{\Delta t}$$

$$W = Fs \cos \theta$$

$$E_k = \frac{1}{2}mv^2$$

$$E_p = mgh$$

$$P = \frac{W}{t} = Fv$$

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

Waves

$$v = f\lambda$$

$$T = \frac{1}{f}$$

$$n = \frac{\sin \theta_1}{\sin \theta_2}$$

$$d \sin \theta = n\lambda \quad (\text{diffraction grating})$$

Ideal Gases

$$pV = nRT$$

$$pV = NkT$$

$$pV = \frac{1}{3}Nm\langle c^2 \rangle$$

$$\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$$

Stefan–Boltzmann & Wien

$$P = \sigma AT^4$$

$$I = \frac{P}{A}$$

$$\lambda_{max}T = 2.898 \times 10^{-3} \text{ m K} \quad (\text{Wien's Law})$$

Electrical Circuits

$$V = IR$$

$$P = IV = I^2R = \frac{V^2}{R}$$

$$Q = It$$

$$E = QV$$

$$\varepsilon = I(R + r)$$

$$R_{series} = R_1 + R_2 + \dots$$

$$\frac{1}{R_{parallel}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

Worked Model

Model: A car accelerates from 8.0 m s^{-1} to 20 m s^{-1} over 6.0 s . Find its acceleration.

G: $u = 8.0 \text{ m s}^{-1}$, $v = 20 \text{ m s}^{-1}$, $t = 6.0 \text{ s}$

U: $a = ?$

E:

$$v = u + at$$

S (rearrange first, then substitute):

$$a = \frac{v - u}{t}$$
$$a = \frac{20 - 8.0}{6.0}$$

S (solve):

$$a = 2.0 \text{ m s}^{-2}$$

Questions

1. A ball of mass 0.40 kg is dropped and hits the ground at 12 m s^{-1} . Calculate its kinetic energy on impact.

Question 1

G: Given.....

U: Unknown.....

E: Equation.....

S: Substitute

S: Solve $\Rightarrow$

2. A trolley of mass 2.0 kg moving at 3.0 m s^{-1} collides and sticks to a stationary trolley of mass 1.0 kg. Find the velocity of the combined trolleys.

Question 2

G: Given.....

U: Unknown.....

E: Equation.....

S: Substitute

S: Solve $\Rightarrow$

3. A sound wave has frequency 440 Hz and wavelength 0.75 m. Calculate its speed.

Question 3

G: Given.....

U: Unknown.....

E: Equation.....

S: Substitute

S: Solve $\Rightarrow$

4. Light passes from air into glass, striking the surface at 40° to the normal and refracting to 25° . Calculate the refractive index of the glass.

Question 4

G: Given.....

U: Unknown.....

E: Equation.....

S: Substitute

S: Solve $\Rightarrow$

5. A diffraction grating with 500 lines per mm is illuminated by light of wavelength 600 nm. Calculate the angle of the first order maximum ($n = 1$).

Question 5

G: Given.....

U: Unknown.....

E: Equation.....

S: Substitute

S: Solve $\Rightarrow$

6. An ideal gas occupies $2.0 \times 10^{-3} \text{ m}^3$ at a pressure of $1.5 \times 10^5 \text{ Pa}$ and temperature 300 K. Calculate the number of moles present.

Question 6

G: Given.....

U: Unknown.....

E: Equation.....

S: Substitute

S: Solve $\Rightarrow$

7. Calculate the mean square speed of gas molecules at 300 K, given the molecular mass is $4.7 \times 10^{-26} \text{ kg}$. (Boltzmann constant $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$)

Question 7

G: Given.....

U: Unknown.....

E: Equation.....

S: Substitute

S: Solve $\Rightarrow$

8. A star radiates as a black body with surface temperature 5800 K and radius 7.0×10^8 m. Calculate the total power output. ($\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2}\text{K}^{-4}$)

Question 8

G: Given.....

U: Unknown

E: Equation.....

S: Substitute

S: Solve $\Rightarrow$

9. A distant star has a peak wavelength of emission at 290 nm. Use Wien's Law to calculate its surface temperature.

Question 9

G: Given.....

U: Unknown

E: Equation.....

S: Substitute

S: Solve $\Rightarrow$

10. A cell of EMF 9.0 V and internal resistance 0.50Ω drives a current of 2.0 A through an external resistor. Calculate the resistance of the external resistor.

Question 10

G: Given.....

U: Unknown

E: Equation.....

S: Substitute

S: Solve $\Rightarrow$